

Frequency Analysis of Pulsar Data

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Abstract

Pulsars are fast spinning, highly magnetic neutron stars, the result of the dying supernovae of $8 - 20 M_{\odot}$ stars. This report uses Fourier analysis and phase binning to analyse the infrared intensity data from a region of the sky believed to contain a pulsar. It is found through Fourier analysis that the data exhibits periodicity with a significance equivalent to 5.091σ , and is very likely not a result of random noise. Phase binning reveals that if there exists a pulsar in the surveyed region, it has a likely rotational period of 33.38 ± 0.41 ms, and a pulse shape defined by a main pulse and a smaller sub-pulse, consistent with common pulsar morphology. Future work should involve a longer duration survey outside the 60 Hz mains region.

1 Introduction to Pulsars

Neutron stars form when a star collapses under its own gravity in a supernova, after running out of sufficient nuclear fuel. To end up as a neutron star, the progenitor star must have enough mass for its gravity to overcome the electron degeneracy pressure which would cause its core to form a white dwarf, $M > 8 M_{\odot}$ [1]. After overcoming electron degeneracy pressure, the atomic nuclei are pressed together enough to convert the protons and electrons into neutrons, forming an extremely closely packed ball of neutrons.

However, too much mass and the gravitational force will exceed the neutron degeneracy pressure, causing total collapse into a black hole. Single stars more massive than $20 - 25 M_{\odot}$ are generally accepted to form black holes on collapse, however it is suggested that with ideal conditions in a binary system, stars of $80M_{\odot}$ may be able to form neutron stars [2].

When a large star dies, its gravitational collapse into a neutron star of radius $\sim 10km$ conserves angular momentum. Stars which before collapse were spinning relatively slowly end up as very fast turning neutron stars, with rotational periods from 10 seconds down to the millisecond range.

Neutron stars also have very strong magnetic fields. The combined factors of very fast rotation, and strong magnetic field, cause the behaviour we observe as pulsars, regular bursts of electromagnetic radiation from the neutron star. The exact production mechanism for the radiation is currently debated, but I will build a quick outline. [3]

The rotating magnetic field acts as a homopolar inductor, generating massive voltages capable of pulling charged particles from the surface of the star or creating charged particles through pair production. These charged particles then move along the magnetic field lines of the star: away from the poles these particles circle back into the star, close to the poles the field lines are open, the particles are accelerated to relativistic speeds as they flow out from the poles on a curved path. Relativistic charged particles emit curvature radiation when they follow a curved path, likely causing the radio emissions from the magnetic poles of these neutron stars. The misaligned rotational and magnetic poles cause this beam to sweep out a circle with each rotation, causing the "lighthouse" effect we observe from Earth. [3]

After years of theoretical construction, suggesting that neutron stars must have large magnetic fields, that they could act as powerful X-ray sources, that a pulsar would explain the activity of the Crab Nebula, the discovery of pulsars was finally announced in 1968. This discovery by Jocelyn Bell Burnell and others was followed a year later with the discovery of the pulsar within the well known Crab Nebula. At just 1000 years old, it is very active, and therefore one of the best studied pulsars. [3] [4]

These discoveries involved finding remarkably stable periodic signals, explainable by pulsars. Frequency analysis techniques such as the Fast Fourier Transform (FFT), converting time-domain data into frequency-domain data, are often used to identify sources of periodic behaviour in the sky [5]. Fourier analysis and phase binning to find pulse shape will be extremely useful for the goal of this report, to analyse 256 data points of infrared intensity from

a region of the sky believed to contain a pulsar, taken every 4 ms.

2 Methodology

The first method of analysis will be a Fourier time series analysis to expose any periodic changes in the intensity, and obtain an approximate value for the frequency of any pulses. Let the signal contain N readings x_t with time interval δt . The Fourier coefficient for each wave number $k = 0, 1, 2, \dots, N - 1$ is then

$$X_k = \frac{1}{N} \sum_{t=0}^{N-1} x_t e^{-2\pi i k t / N}$$

The power of each component is then given by $P_k = |X_k|^2$, and the k values correspond to the frequencies $f = k/N\delta t$. A power spectrum of P_k against f can then be plotted. Since the sampling is discrete and not continuous, the spectrum will be symmetrical around half the sampling frequency $1/2\delta t$. I checked this implementation using a test function with 256 points: a combination of 20 Hz, 50 Hz, and 80 Hz sine waves, with the 20 Hz wave having twice the amplitude in figure 1.

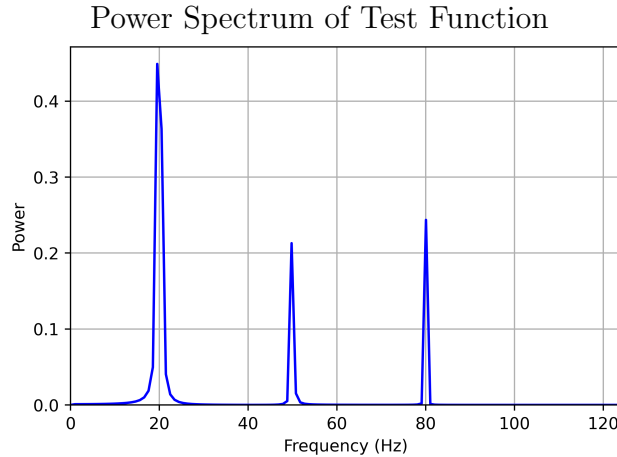


Figure 1: This figure shows the power spectrum obtained by running a test function through the Fourier analysis implementation detailed above, showing a frequency spike at 20 Hz, 50 Hz, and 80 Hz, indicating a working Fourier implementation.

After finding the approximate frequencies of periodic behaviour using Fourier analysis, the second method of analysis will be phase binning. Phase binning involves splitting the data into sections the size of a period of the pulse found from the power spectrum. By creating bins inside that period and assigning points into each bin, you can find the average shape of each period.

Let t be the time at each data point and T be the period we are checking. The phase from 0 to 1 of each point is given by $\phi = \frac{t \bmod T}{T}$ and sorted into equal size bins. Each sample's intensity can be summed or averaged across each bin, then plotted against time in phase to get the shape of the periodic behaviour.

It is important to note that a smaller period (larger frequency) has fewer sampled points per cycle. Over many cycles, these points create clusters of samples within the cycle, so ideally each cycle should have at least 1 sample per bin so that every bin gets enough samples to create an accurate average intensity. Therefore, the number of bins should be chosen based on how many sample points occur in a single period.

This process can be repeated over frequency intervals around the peaks we are interested in, and standard deviation of average intensity across bins in the phase can be used as a measure of how well that specific frequency captures the periodic behaviour of the pulse, with higher standard deviations indicating a better frequency. In this way, a more accurate pulsar period can be obtained than from the initial Fourier analysis.

3 Fourier Analysis

Running the data through the Fourier analysis method detailed in the last section results in figure 2. This plot shows a fundamental frequency f_1 at around 30 Hz, then a higher power spike f_2 at around 60 Hz. As will be explored in the next section and figure 6, the increased power of the f_2 spike is likely explained by the f_1 peak corresponding to the main pulse of the pulsar, but containing a smaller sub-pulse, also known as an interpulse, around halfway between each main pulse, which is combined into one peak when analysing at 60 Hz.

The other peaks visible on the graph also correspond to harmonics of f_1 : f_3 at 90 Hz, f_4 at 120 Hz. The spike at 100 Hz is likely the f_5 peak; as previously mentioned, the discrete form of our data means that the graph

is reflected around half the sample frequency, 125 Hz for our graph, so the normal 150 Hz position of the f_5 peak is reflected across 125 Hz.

Power Spectrum of Infrared Intensity from Pulsar Candidate Data

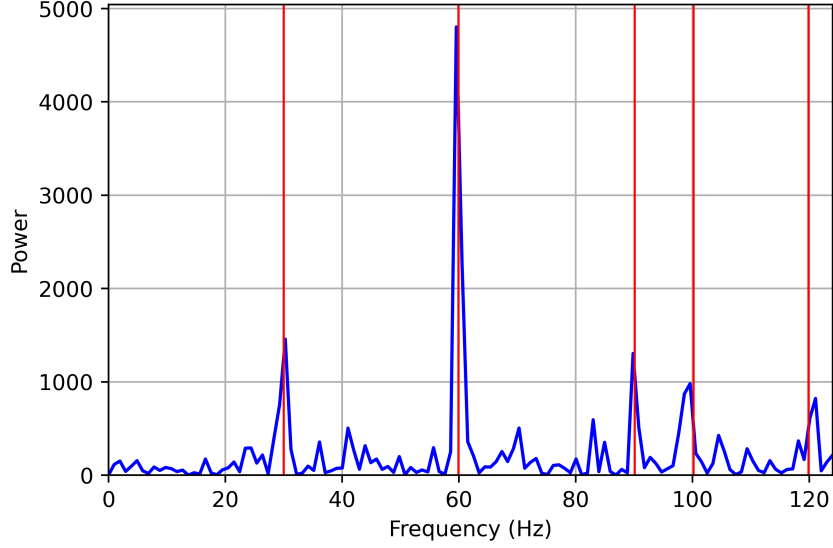


Figure 2: This figure shows the power spectrum of the infrared intensity data, labelled with red lines from left to right at $f_1 = 30$ Hz, $f_2 = 60$ Hz, $f_3 = 90$ Hz, $f_5 = 100$ Hz, $f_4 = 120$ Hz.

The graph very clearly displays that the data shows periodic behaviour, likely indicating the presence of a pulsar in the surveyed location. By randomly generating many other datasets with the same standard deviation as our data, and running them through the Fourier transform, then comparing the distribution of the largest randomly generated peak to our own data, we can quantify the likelihood that the observed periodicity occurred from randomness.

Figure 3 shows the distribution of the largest power spike in 1000 samples of random noise generated with the same standard deviation as our collected data. Modelling the curve as a Gumbel distribution, we can compare the area under the curve to find that the false alarm probability of our data is $1.780 \cdot 10^{-7}$, equivalent to 5.091σ . We can therefore be quite confident that this is real observed periodic behaviour and not a result of random noise.

Probability Density of Largest Power Spike in Random Noise

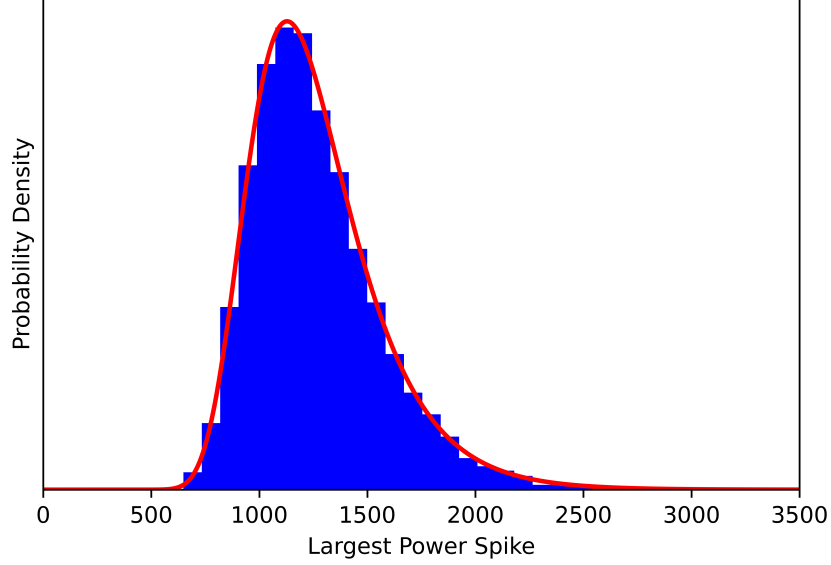


Figure 3: This figure shows a histogram of the largest power spike found in random samples of data with a standard deviation equal to our observational data. In red is the Gumbel distribution used to find the probability of a ~ 4900 power peak from random noise alone.

4 Phase Binning

The first undertaking using phase binning is to find a more accurate number for the fundamental frequency of our f_1 spike from the previous section, which was estimated to be at around 30 Hz. As explained, this involves phase binning with many periods in the range around 30 Hz. In order to have at least 1 sample per cycle per bin, see that $256/30 = 8.533$ samples per period, and choose 8 bins as a good number.

Figure 4 shows the results of the phase binning test from 29 – 30 Hz. Although a standard deviation maximum is seen at exactly 30 Hz, a better result is obtained by calculating the weighted mean and standard deviation of the peak with the standard calculations, using earlier standard deviation as the measure of periodicity for the weighting. This process resulted in a fundamental frequency estimate of 29.96 ± 0.36 Hz. The same method was used to find the position of the f_2 peak, resulting in a value of 59.96 ± 0.61 Hz. Corresponding to a period of 33.38 ± 0.41 ms and 16.69 ± 0.17 ms respectively.

Standard Deviation in Phase Binned Points Across A Range of Frequencies

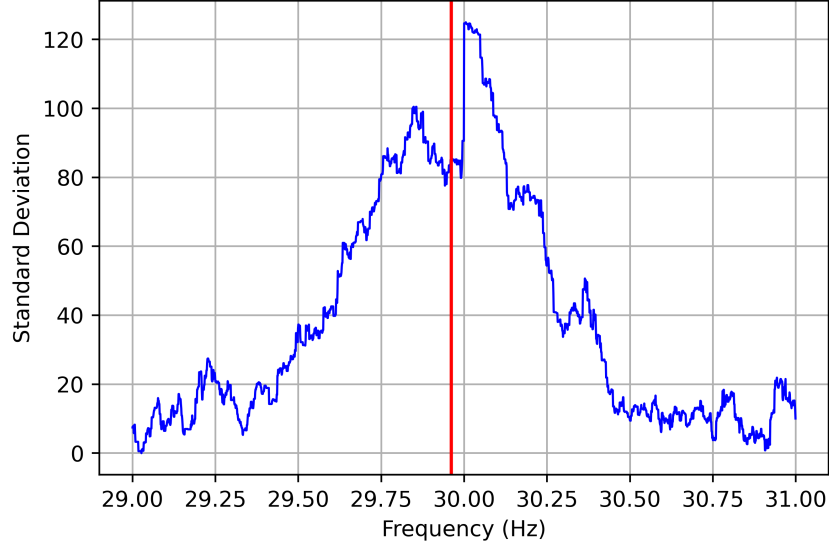


Figure 4: The standard deviation in the results from phase binning the data at different frequencies. Larger standard deviations between points indicate a better captured periodic behaviour in that frequency. There is a maximum standard deviation at almost exactly 30 Hz, but the mean centre of the peak is indicated by the red line at 29.96 ± 0.36 Hz

Now that we have accurate values for the important peaks, we can use the phase binning algorithm to find the average shape of the pulses at these periodicities. In order to get an initial idea about the pulse shape I generated a pulse stack graph at 29.96 Hz, and a phase binned average pulse at 15 Hz showing 2 periods of the pulse, displayed in figure 5.

The phase stack shows each pulse stacked on top of one another, with a 30 Hz frequency, older pulses at the top. You can see a consistent main pulse on the right side of the period, and a less consistent smaller pulse on the left side of the period. This is the first suggestion that 30 Hz is the main frequency of the pulsar candidate, with a smaller sub-pulse causing the large 60 Hz power spike.

For the for the graph over 2 periods, 16 bins gives at least 1 sample per bin per cycle, I found that using 32 bins increases resolution while not losing the shape of the graph. Checking the number of samples per bin, using 32 still gives enough points per bin, with no bin falling far below any other.

Visualisations of Pulsar Candidate Pulse Shape

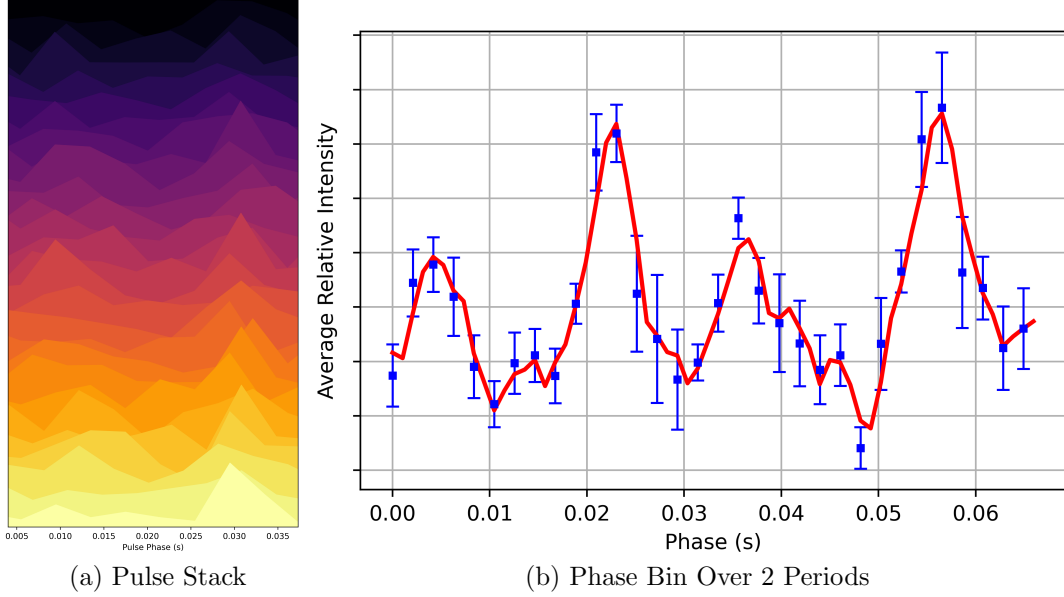


Figure 5: Displaying the pulse shape of the pulsar candidate. The left pulse stack shows each 33.38 ms period stacked on top of each other, the right phase bin graph shows the average pulse shape over a 2 period time interval. In both visualisations you can see the pattern of alternating large and smaller pulses.

The crucial point is that the shape of the graph overall is unaffected by this doubling, but starts to become less descriptive with more bins.

The red line is a smoothed 3-point average over a 64 bin phase binning, it is not for analytical purposes but to more easily show the overall structure of the pulses, which seems to be a pattern of large pulses with a smaller pulse halfway between them.

Figure 6 shows the phase binned graphs for a single period, 33.38 ms, and for half a period, 16.69 ms. Again, after generating these graphs with 8 and 4 bins respectively, guaranteeing at least 1 sample point per bin per cycle, I used 16 and 8 bins for the graphs due to no loss of overall graph shape plus increased resolution. The red line is for easier visualisation as before.

From these graphs it is possible to conclude, as earlier mentioned, that the 60 Hz spike from figure 2 is taller because it combines the main pulse with the smaller sub-pulse. The 60 Hz pulse also has an abnormally wide

pulse shape compared to its period for a pulsar, whose main pulse usually only takes up a small part of the period, as in the left graph. It is therefore very likely that, if the targeted region of space contains a pulsar, its period is 33.38 ± 0.41 ms rather than 16.69 ± 0.17 ms.

Comparison of Phase Binning of 33 ms and 17 ms Pulsar Periods

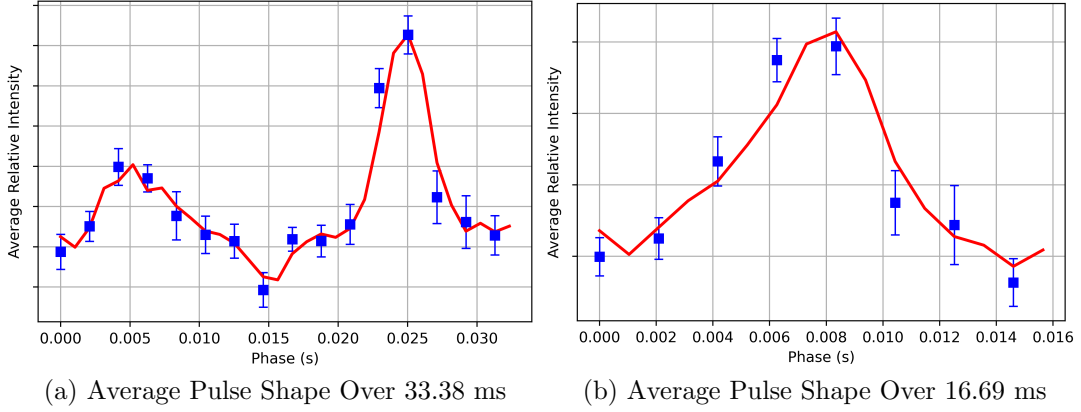


Figure 6: The two possibilities for pulse fundamental frequency compared, found through phase binning, the left corresponds to the 30 Hz spike and the right to the 60 Hz spike in figure 2. The 60 Hz spike therefore corresponds to the combining of both pulses from the 30 Hz graph into a single pulse.

5 Discussion

Fourier analysis and phase binning both identified periodic behaviour in the infrared intensity consistent with a pulsar in the region, with a rotational frequency of 29.96 ± 0.36 Hz, corresponding to a rotational period of 33.38 ± 0.41 ms. Harmonics of the rotational period were observed up to f_5 , explaining every major peak in the frequency domain graph figure 2. The second harmonic f_2 showed a very strong peak, likely due to the combining of the main pulse and sub-pulse of the signal into a single peak, rather than two peaks as in the f_1 spike.

Through phase binning, the pulse profile of the signal was shown to consist of large pulses around every 33 ms, with small sub-pulses halfway between the large pulses. The pulse profile is also consistent with the existence of a 33 ms pulsar within the surveyed region, with the main pulse taking up

just a small part of the total period, and the slightly inconsistent, smaller, sub-pulse. Both of these features are common in pulsar signals. [3]

The Fourier transformation was used mainly as an initial probe into any periodic behaviour of the signal and the positions of the periodicities. The false alarm probability of finding the scale of periodicity in random noise was found with a Monte Carlo method to be $1.780 \cdot 10^{-7}$, equivalent to 5.091σ . It is therefore very likely that the periodicity comes from some source, and is not a random noise artefact.

The phase binning used a number of bins needed to have at least 1 sample point per bin per cycle for all quantitative findings, ensuring an accurate mean and uncertainty in the final values for rotational period. For qualitative displays, when showing the shape of the pulse, twice the bins were used to increase resolution while ensuring the overall pulse shape did not change.

Although noise has been ruled out as a source of periodicity in the data, it may be possible that the 60 Hz spike is produced through electrical interference. The mains frequency of some regions, including the United States, is 60 Hz. The 30 Hz spike could then be formed as a sub-harmonic, or through random noise, which is more likely than for the more prominent 60 Hz peak. The existence of the 30 and 90 Hz peaks, as well as the pulse/sub-pulse shape of the 30 Hz peak, makes this possibility unlikely.

In order to be more certain of the existence of a 33 ms pulsar in this region, a future survey should check the infrared intensity from this region for a longer time, our data only spans 1.024 s, which would allow a higher resolution, lower noise investigation into the pulse profile. A shorter sampling interval would also provide more data, while also increasing the possible frequency range probed by the power spectrum graph. A survey from outside the United States could rule out the unlikely but still possible case of the 60 Hz spike being caused by electrical interference, less mains interference may also reduce noise.

In conclusion, Fourier analysis and phase binning have been employed to say with quite high certainty that there is periodic behaviour within the data, and that it is likely to be the result of a 33 ms pulsar within the surveyed region, with a consistent main pulse and a less consistent sub-pulse, best visible in figure 5.

References

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